|NOT MEASUREMENT SENSITIVE|

MIL-D-29227A
28 October 1988
SUPERSEDING
MIL-D-29227(YD)
8 August 1983

MILITARY SPECIFICATION

DITCHING MACHINE, WHEEL TYPE TRENCHER, CRAWLER-MOUNTED, DIESEL-ENGINE-DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a commercial, diesel-engine-driven, crawler-mounted, wheel trenching type ditching machine.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

AMSC N/A

|FSC 3805|

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:22Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronics, and Associated Equipment).
MIL-T-3351 - Tractor, Full-Tracked, Low Speed; Tractor Wheeled, Agricultural; and Tractor Wheeled, Industrial, and Their Attachments, Packaging of.

STANDARDS

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

(Unless otherwise indicated, copies of Military specifications and standards are available from the Naval Publications and Forms Center (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J534 - Fittings, Lubrication.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:22Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Description. The ditching machine shall be a diesel-engine-driven, crawler mounted, wheel trenching type, used for general purpose ditching, excavating for building foundations, drainage, and pipeline ditches.

3.2 Standard commercial product. The ditching machine shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the ditching machine being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection (see 4.2.1 and 6.3).

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Safety. All rotating or reciprocating parts and all parts subject to high operational temperatures, that are of such a nature or are so located as to be or become a hazard to the safety of the operating personnel, shall be insulated, enclosed, or guarded to the extent necessary to eliminate the hazard.

3.7 Operational requirements. The ditching machine shall perform as specified herein in any ambient temperature from plus 10 degrees Fahrenheit (°F) to plus 110°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:22Z
Check the source to verify that this is the current version before use.

3.8 Shipping weight. The shipping weight of the ditching machine shall be not less than 16,000 pounds (lb) and not more than 26,000 lb.

3.9 Performance. The ditching machine shall be capable of digging into various earth formations, a ditch up to 24 inches in width, and (a) not less than 5 feet 10 inches deep in form clay soil at a digging rate of not less than 5 feet per minute (fpm), or (b) not less than 5 feet 10 inches deep in clay-gravel soil at a digging rate of not less than 4 fpm. The crowding speed shall be selectively controlled by the operator. Walls and bottom of the cut shall be clean and true to line and grade. The discharge conveyor shall be capable of depositing spoil in windrows at either side of the excavation and at a distance sufficient to prevent re-entry of spoil into the trench.

3.9.1 Travel speeds. The ditching machine shall have not less than three travel speeds forward and one reverse or an infinitely variable speed hydrostatic drive. The highest forward travel speed shall be not less than 2.0 miles per hour (mph).

3.9.2 Digging speeds. The ditching machine shall have not less than five digging (crowding) speeds ranging from not greater than 3 fpm to not less than 6 fpm or having variable digging (crowding) speeds from 0 to not less than 30 fpm at engine governed speed. The crowding or digging speeds shall be so selective as to provide maximum efficiency at all times. A hydrostatic-drive that will meet required digging speeds is acceptable.

3.10 Design and construction.

3.10.1 Wheel assembly. The diameter of the wheel assembly, measured to the tips of the bucket teeth, shall be not less than 115 inches.

3.10.2 Buckets. The ditching machine shall be equipped with buckets and teeth to cut trenches, one at a width of not less than 18 inches, and one at a width of not less than 24 inches. The cutting width requirements may be met with bolted-on side cutters. The teeth shall be of the quick interchangeable type. Replacement of teeth shall be accomplished without the use of special tools.

3.10.3 Bucket cleaner. Manufacturer's standard type bucket cleaner shall be furnished.

3.10.4 Spoil conveyor. The spoil conveyor shall be individually controlled, dischargeable to either side, from the operator's seat, and belt speed so controlled as to provide the correct operational speed to give maximum or minimum discharge. The spoil conveyor positioning shall be hydraulically powered.

3.10.4.1 Spoil conveyor belt. The spoil conveyor belt shall be of adequate width to handle the spoil while digging at maximum depth and maximum crowding speed. The belt drive shall be hydrostatic or hydraulically driven.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:22Z
Check the source to verify that this is the current version before use.

3.10.5 Framework. The digging boom, spoil conveyor, and powerplant shall be mounted in a framework fabricated from steel and shall be bolted, riveted, welded, or a combination thereof, reinforced and braced to maintain proper alignment of all components through all operations.

3.10.6 Crawler. The crawler shall have steel semigrouser, or grouser type track shoes. The track shoes shall be not less than 16 inches in width. Provisions shall be made to adjust for slack due to wear.

3.10.7 Engine. The ditching machine shall be powered by a diesel engine. The ditching machine shall be powered by the manufacturer's standard commercial engine normally furnished to the commercial market for this size ditching machine. The power and speed required of the engine to meet the performance requirement of the ditching machine shall be not greater than the continuous horsepower values at which the engine is rated. The engine shall be furnished with accessories and an electrical cranking system normally furnished on the manufacturer's commercial standard ditching machine.

3.10.7.1 Dry charged batteries. Batteries furnished shall be dry charged, without electrolyte, including sealed caps to prevent the intrusion of atmospheric moisture.

3.10.8 Hydraulic system. A complete hydraulic system, including a pump, hydraulic rams, reservoir, pressure relief valve, and all necessary lines and controls, shall be provided to perform all required hydraulic operations. The pump shall be driven by the ditching machine engine and shall have sufficient capacity to energize all hydraulically powered components operating simultaneously.

3.10.9 Controls. All controls shall be located within easy reach of the operator and shall be clearly marked as to use and function.

3.11 Lifting and tiedown attachments. The ditching machine shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the ditching machine. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the ditching machine on the carrier when shipped.

3.12 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:22Z
Check the source to verify that this is the current version before use.

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.13 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 pound-force per square inch (psi) or higher, will damage grease seals or other parts, a warning label shall be affixed to the equipment in a conspicuous location near concerned lubrication point. The unit shall be lubricated as specified in section 5.

3.14 Cleaning, treatment, and painting. Surfaces normally painted shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.15 Identification plate. An identification plate will be furnished by the contracting officer for each ditching machine. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each ditching machine in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.16 Instruction plates. The ditching machine shall be equipped with instruction plates located as to be visible from operator's platform, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch.

3.17 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:22Z
Check the source to verify that this is the current version before use.

3.18 Workmanship.

3.18.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. All bends shall be made by controlled means to insure uniformity of size and shape.

3.18.2 Bolted connections. Boltholes shall be punched or drilled within manufacturer's design tolerances and shall have the burrs removed. Washers or lockwashers shall be provided, and all bolts, nuts, and screws shall be tight.

3.18.3 Riveted connections. Rivet holes shall be punched or drilled within manufacturer's design tolerances and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.18.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.18.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:22Z
Check the source to verify that this is the current version before use.

products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practic to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one ditching machine when a first article is required (see 3.3 and 6.2). This inspection shall include the examination of 4.3, the tests of 4.4 and, when specified, the preproduction pack inspection of 4.6 (see 4.6 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.6.

4.3 Examination. Each ditching machine shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 First article tests. The ditching machine shall be subjected to the tests specified in 4.4.1 through 4.4.3 to determine ruggedness of construction and satisfactory performance. Failure to pass any phase of these tests shall constitute cause for rejection.

4.4.1 Digging tests. The digging tests shall be performed in one or more of the formations specified in 3.9. This test is to provide an accumulative digging time of not less than 4 hours, varying in depth from 4 feet to 5 feet 10 inches in order to test the production capability as required in 3.9. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:22Z
Check the source to verify that this is the current version before use.

spoil discharge shall be alternated from one side to the other not less than five times during the digging tests. The boom shall not be elevated during the digging tests, except to remove or pass over obstacles extending beyond the width of the trench or beyond the depth capacity of the ditching machine.

4.4.2 Travel test. The ditching machine shall be operated in each travel speed specified in 3.9.1. The forward travel speed shall be measured to determine conformance to 3.9.1.

4.4.3 Lifting and tying down attachments test. When lifting and tying down attachments are required, the ditching machine shall be tested to determine conformance to 3.11.

4.5 Production unit operation test. Each production ditching machine produced in fulfillment of a Government contract or order shall be completely assembled, adjusted, lubricated, or otherwise serviced for operation. The engine shall be started and subjected to a warmup period as recommended by the manufacturer. The ditching machine shall be given a run-in test and all controls operated a sufficient number of times to ascertain that the components and mechanisms actuated by the controls operate promptly, fully, and without restriction or malfunction. Failure to pass any phase of this test shall be cause for rejection.

4.6 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-T-3351. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-T-3351, with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The ditching machine covered by this specification is intended for use in general purpose ditching, excavating for building foundations, drainage, and pipeline ditches at military installations.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:22Z
Check the source to verify that this is the current version before use.

c. When first article and inspection sample is required (see 3.3 and 4.2.1).
d. When treatment for fungus resistance is required (see 3.12).
e. Means for lubrication, if other than as specified (see 3.13).
f. Color of finish coat (see 3.14).
g. When a preproduction pack inspection is required. Time frame for preproduction pack inspection (see 4.2.1 and 4.6).
h. Level of preservation of level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DOD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DOD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item(s) should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first article. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Subject term (key word) listing.

Crawler mounted
Ditching machine
Wheel type trencher

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Review Activity:
DLA - CS

Preparing activity:
Navy - YD

Project 3805-0130

☆U.S. GOVERNMENT PRINTING OFFICE: 1989 - 604-033/10175

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:22Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer (Code 156)
Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Code 156
Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:22Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCU[illegible]MPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-29227A	DITCHING MACHINE, WHEEL TYPE TRENCHER, CRAWLER-MOUNTED, DIESEL-ENGINE-DRIVEN

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:22Z
Check the source to verify that this is the current version before use.